
```

title:... "Time-Off... Micro... ervice...-... How... It... Wa..... Implemented"
author:... "ExampleHR... En... ineerin..."
date:... "2026-05-12"
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```

#... Tech... Stack

... Concern...	... Choice...
... Framework...	... Ne... tJS... (Node.j.....+... TypeScript..... trict)...
... Databa... e...	... SQLite... via... TypeORM...+... better-... qlite3,... WAL...
... HTTP... client...	... @ne... tj..... axio.....+... axio...-retry...
... Validation...	... cla.....-validator...+... cla.....-tran... former...
... Rate... limitin.....	... @ne... tj..... throttler...
... API... documentation...	... @ne... tj..... wa..... er... (OpenAPI... 3.0,... Swa.....
... Te... t.....	... Je... t...+... Superte... t...+...nock...
... Mock... HCM...	... Standalone... Ne... tJS... app... (port... 3001)...

#... Project... Structure

```

...rc...
|--...confi.....#...Typed...env...var.....wit
|--...common...
|.....|--...concurrency.....#...retryOnOptimi...ticLock...+...Optimi...t
|.....|--...filter.....#...Global...HTTP...exception...>
|.....|--...uard.....#...JwtAuthGuard...(...tub)...
|.....\--...interceptor.....#...Re...pon...eInterceptor...+...Lo.....in
|--...databa...e...
|.....|--...tran...action.helper.t.....#...withImmediateTran...action...(BEGIN
|.....\--...mi...ration.....#...Sin...le...S
\--...module.....
.....|--...balance.....#...Balance...read...,...HCM...rei
.....|--...time-off.....#...State...machine,...balance...al
.....\--...hcm-...ync.....#...Batch.....in...le.....ync...

mock-hcm.....#...Standalone...HCM...mock...
te...t...
|--...unit.....#...Fully...mocked.....ervi
|--...inte...ration.....#...Superte...t...+...nock,...in-memory
\--...e2e.....#...Live...mock...HCM,...

```

Swa.....er...UI...i.....erved...at.....api...doc.....(OpenAPI...JSON...at.....api...doc...-

#...The...Two...Critical...Helper...

Everythin.....el...e...in...the...implementation...depend.....on...the...e...two...utilitie....

##...withImmediateTran...action

```
export...a...ync...function...withImmediateTran...action<T>(  
  ....dataSource:...DataSource,  
  ....fn:... (mana...er:...EntityMana...er)...=>...Promi...e<T>,  
):...Promi...e<T>...{  
  ....con...t...qr...=...dataSource.createQueryRunner()...  
  ....await...qr.connect()...  
  ....await...qr.query(...BEGIN...IMMEDIATE...)...  
  .....Patch...TypeORM.....tate.....o...ne...ted.....ave()...u...e.....SAVEPOINT...  
  ....(qr...a.....any)...i...Tran...actionActive...=...true...  
  ....(qr...a.....any).tran...actionDepth...=...1...  
  ....try...{  
  .....con...t...re...ult...=...await...fn(qr.mana...er)...  
  .....await...qr.query(...COMMIT...)...  
  .....return...re...ult...  
  .....}...catch...(err)...{  
  .....await...qr.query(...ROLLBACK...)...  
  .....throw...err...  
  .....}...finally...{  
  .....await...qr.relea...e()...  
  .....}  
}
```

Why...the...TypeORM...patch?...When...TypeORM.....EntityPer...i...tExecutor...i.....called
...which...SQLite...reject.....becau...e...we...re...already...in...ide...BEGIN...IMMEDIATE...Settin...

##...retryOnOptimi...ticLock

```
export...a...ync...function...retryOnOptimi...ticLock<T>(  
  ....fn:...()...=>...Promi...e<T>,  
  ....maxAttempt.....=...5,  
):...Promi...e<T>...{  
  ....for...(let...attempt...=...1.....attempt...<=...maxAttempt.....attempt++)...{  
  .....try...{...return...await...fn().....}  
  .....catch...(err)...{  
  .....if...(! (err...in...tanceof...Optimi...ticLockError))...throw...err...  
  .....await.....leep(ba...eDelay...*...attempt...+...jitter)...  
  .....}  
  .....}  
  .....throw...la...tErr...  
}
```

The...retry...loop...wrap.....the...*entire*...tran...action...body...-
 ...includin.....re-readin.....the...balance...row...-.....o...each...retry...always.....ee.....fre...h

#...The...Balance...Repo...itory

All.....ix...mutation...method.....follow...the.....ame...pattern:...a...raw...query-
 builder...UPDATE...with...an...explicit...ver...ion...check,...throwin.....Optimi...ticLockError...if.

```

.....Example:...re...ervin.....day.....when...a...reque...t...i.....created
a...ync...applyHcmRefre...hAndIncrementPendin...(
.....mana...er:...EntityMana...er,
.....balance:...LeaveBalance,
.....newHcmBalance:...number,
.....day...:...number,
):...Promi...e<void>...{
.....con...t...re...ult...=...await...mana...er
.....createQueryBuilder()
.....update(LeaveBalance)
.....et({
.....hcmBalance:...newHcmBalance,
.....pendin...Day...:...()...=>...`pendin..._day.....+...${day...}` ,
.....ver...ion:...:...()...=>.....ver...ion...+...1...,
.....})
.....where(...id...=...:id...AND...ver...ion...=...:ver...ion...,
.....{...id:...balance.id,...ver...ion:...balance.ver...ion...})
.....execute()...

.....if...(re...ult.affected...===...0)...throw...new...Optimi...ticLockError()...
}

```

Why...one...method...that...both...refre...he.....HCM...balance...AND...increment....

#...The...createReque...t...Flow...(The...Harde...t...Method)

```

1...Check...idempotency...key...in...DB...(out...ide...txn...-...UNIQUE...con...straint...ha
2...Call...HCM.....etBalance...(out...ide...txn...-...don...t...hold...write...lock...duri
.....\-...If...HCM...down:...u...e...-1.....entinel...(fall...back...to...local.....ha
3...retryOnOptimi...ticLock:
.....\-...withImmediateTran...action:
.....a....Load...balance...row
.....b....LOCAL...check:...hcmBalance...-...pendin...Day.....>=...day.....
.....c....AUTHORITATIVE...check:...hcmAvailable...-...pendin...Day.....>=...da
.....d....applyHcmRefre...hAndIncrementPendin.....(optimi...tic...lock)
.....\-...throw.....Optimi...ticLockError.....>.....retry...from...
.....e....INSERT...time_off_reque...t
.....\-...UNIQUE...violation...on...idempotency_key.....>.....return

```

```

.....f.....INSERT...balance_audit_lo.....(...ame...tran...action)
4....Return...201

```

Why...two...balance...check.....(b...and...c)? Check... (b)...u...e.....the...local.....hadow
...it.....fa...t...and...catche.....the...obviou.....ca...e....Check... (c)...u...e.....the...fre...h...
...it...catche.....the...ca...e...where...the...HCM...balance...decrea...ed...externally... (e.....,...a...
trip.

#...The...approveReque...t...Flow

```

1....Load...reque...t,...a.....ert...PENDING
2....Call...HCM...deductBalance...(out...ide...txn)
.....|-...422...(in...ufficient)...>...throw...409,...DB...unchan...ed
.....\-.5xx.....timeout...>...throw...502,...DB...unchan...ed
3....retryOnOptimi...ticLock:
.....\-.withImmediateTran...action:
.....a....Re-read...reque...t...(...uard...a...ain...t...double-approve)
.....b....Load...balance...row
.....c....applyApproval:...hcm_balance...-=...day...,...pendin..._day.....-=...
.....d....UPDATE...reque...t.....tatu.....to...APPROVED...+...hcm_reference_...
.....e....INSERT...balance_audit_lo...
4....Return...200

```

Why...re-read...the...reque...t...in...ide...the...tran...action? The...mana...er...could...d
click... “Approve”...The...fir...t...call.....ucceed.....and...commit.....The.....econd...call...re-
read.....the...reque...t,...ee.....it.....already... APPROVED,...and...return.....ilently...-
...idempotent...without...any...HCM...re-call.

#...Batch...Sync...Reconciliation

Each...balance...entry...in...the...HCM...batch...payload...i.....proce.....ed...in...it.....own...i...

For...each...entry:

```

.....1....Up...ert...leave_balance...with...new...hcm_balance...(INSERT...or...UPDATE...+...
.....2....Recompute...pendin..._from_db...=...SUM(day...)...WHERE.....tatu...=...PENDING...
.....3....If...(new_hcm_balance...-...pendin..._from_db)...<...0:
.....>...LOG...WARN...di...crepancy
.....>...do...NOT...auto-cancel...reque...t.....(human...mu...t...review...
.....4....INSERT...balance_audit_lo.....(...ource=...BATCH_SYNC...)

```

Proce.....in.....entrie.....individually...(not...in...one.....iant...tran...action)...mean.....a.....in

#...Mock...HCM...Server

The...mock...HCM...i.....a...fully...functional...Ne...tJS.....erver...with:

-...In-memory...Map<...trin...,...number>...for...balance...-...Real...422...Unproce.....able...I

-...POST.....hcm...debu.....anniver...ary...that...increment.....balance...then...a...ynchronou...
...provin.....the...webhook...inte...ration...work... -...POST.....hcm...debu.....re...et...for...t

#...Te...t...Re...ult...

...Suite...	...File.....	...Te...t.....	...Covera...e...
...Unit...	...6...	...67...	...-...
...Inte...ration...	...6...	...30...	...-...
...E2E...	...4...	...8...	...-...
...Total...	...15...	...105...	...-...

...Metric...	...Score...	...Thre...hold...
...Statement.....	...92.1%...	...80%...OK...
...Branche.....	...80.3%...	...75%...OK...
...Function.....	...92.3%...	...80%...OK...
...Line.....	...91.7%...	...80%...OK...

All...5...mandatory...re...re.....ion.....cenario.....have...dedicated,...named...te...t...file...

#...Notable...Implementation...Di...coverie...

The...SAVEPOINT...problem...TypeORM.....mana...er...ave()...internally...check.....i...
...which...SQLite...reject.....Patchin.....qr.i...Tran...actionActive...=...true...after...the...ra...
world...TypeORM...+...SQLite.....otcha...not...documented...anywhere.

SQLite...WAL...+...optimi...tic...lockin.....SQLite...in...WAL...mode...allow.....concurrent...
...the...timeout.....ive.....the...optimi...tic-lock...retry...loop...time...to...work.